



Quantification of poly(ethylene terephthalate) in environmental samples after methanolysis via gas chromatography–mass spectrometry

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Abstract

Quantification of the polyester poly(ethylene terephthalate) (PET) in environmental samples is a particular challenge. Due to strong matrix effects by inorganic compounds, thermoanalytical methods are not recommendable for a precise quantification of PET in complex environmental matrices. It was shown that depolymerization followed by determination of chemolysis products is a good alternative. In this study, we developed a quantification method for PET based on methanolysis to terephthalic acid dimethyl ester using sodium methoxide as a catalyst, and subsequent determination via GC–MS. With poly(ethylene terephthalate- d_4), we introduce a new internal standard covering the whole analytical process. Satisfactory detection and quantifications limits (1 and 4 $\mu\text{g g}^{-1}$) as well as recoveries of 87–117% were achieved. Tests with various PET-free natural compounds exhibited no interfering matrix effects. The newly developed method was applied for MP quantification in a variety of environmental samples such as sediments, sewage sludge, indoor dust, and water. In all these matrices, PET was present. Highest concentrations were detected in indoor dust with up to 57 mg/g. In bottled water, PET concentrations were detected as high as 463 ng/L. The described depolymerization method offers a straightforward approach for a reliable quantification of PET in complex environmental matrices suitable for routine analysis.

Keywords Microplastics · Polyesters · Sediment · Sewage sludge · Indoor dust · Bottled water

Introduction

The quantification of microplastics (MP) in environmental samples is still a challenging task. As synthetic polymers are a heterogenic compound class encompassing a large variety of different chemical structures and functional groups, there is no universal method for their quantification. Instead, due to their different chemical and physical properties (e.g., density, solubility, behavior during thermal degradation), specific methods are needed for different polymer types. Thermoanalytical methods, such as pyrolysis–gas chromatography–mass spectrometry (Py–GC–MS), have proven to

be suitable for the quantification of polyethylene, polypropylene, polystyrene, polyurethanes, polycarbonates, as well as tire wear particles in environmental samples [1–5]. Other polymers such as poly(vinyl chloride) deliver no specific thermolysis products for a reliable quantification in complex environmental samples. A combustion–ion chromatography method using chloride for quantification has been described as a promising alternative [6].

Quantification of the polyester poly(ethylene terephthalate) (PET) in environmental samples, however, is a particular challenge, due to strong matrix effects by inorganic compounds during thermolysis leading to changes in the distribution of pyrolysis products, reactions with other analytes, or losses of intensity due to decomposition of the specific markers used for identification and quantification [7]. Thus, thermoanalytical methods are not recommended for direct quantification of PET in complex environmental and biological matrices [7]. These challenges can be approached by an extensive sample preparation and the use of tetramethylammonium hydroxide (TMAH) as a methylation reagent [2, 8, 9].

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Alternative approaches based on chemolysis of the polymers prior to detection have been developed. These methods are based on the depolymerization of PET by chemical reactions forming terephthalic acid (hydrolysis) [10–12] or the corresponding 2-hydroxyethyl esters (glycolysis) [13]. In the cited studies, determination of the chemolysis products was performed by liquid chromatography using UV or mass selective detection. However, depolymerization was realized under harsh conditions (elevated pH and temperature). A precise reaction control is needed to prevent the formation of unintended side products. An alternative depolymerization approach is reported to be transesterification to terephthalic acid dimethyl ester (dimethyl terephthalate, DMT) as it is used in chemical recycling of PET [14, 15]. Under mild conditions (ambient temperature) and with the help of a suitable catalyst such as K_2CO_3 [14] or sodium methoxide (CH_3ONa) [15], PET waste can be quantitatively transformed into DMT, which is an educt for the synthesis of new PET. These straightforward methanolysis approaches have the potential to be adapted as a first step in a quantification method for PET in environmental samples.

In this paper, we present the development of a new quantification method for PET based on catalytic methanolysis to DMT and subsequent detection via GC–MS. With poly(ethylene terephthalate- d_4) (PET- d_4), we are introducing a new internal standard (IS) covering the whole analytical procedure. The described method has been designed for a high-throughput analysis of PET at low concentrations in complex matrices such as sediments, soil, and sewage sludge.

Material and methods

PET reference mixtures

PET ($M_n = 58,700 \text{ g mol}^{-1}$, $D = 1.96$) was purchased from PSS-Polymer (Mainz, Germany) and ground in a cryo-mill (Retsch, Haan, Germany) to a fine powder. Exact 100 mg of PET powder was mixed with 9.9 g calcined sea sand (700 °C, 3 h) to obtain 10 g of polymer/sand mixture with a PET concentration of 10 mg g^{-1} . This mixture was homogenized for 20 min in a planet mill (Fritsch, Idar-Oberstein, Germany). To obtain calibration curves, the stock mixture was serially diluted with sea sand (Sigma-Aldrich, Schnelldorf, Germany) by diluting 1–2 g mixture with an adequate quantity of sea sand to reach the desired polymer concentrations for each calibration point. The mixture was homogenized in a planet mill after each dilution step. As IS, a 1 mg g^{-1} mixture of PET- d_4 ($M_n = 40,000 \text{ g mol}^{-1}$, broad dispersity; PolymerSource, Montreal, Canada) in calcined sea sand was prepared and homogenized in a cryo-mill.

Methanolysis using K_2CO_3

An equimolar mixture of dichloromethane (DCM) and methanol (MeOH), both obtained from Merck (Darmstadt, Germany), was prepared by mixing 61 mL of DCM with 39 mL of MeOH (per 100 mL of mixture). Exact 50 mg PET/sand mixture was weighed into a reaction vessel and 5 mL DCM/MeOH mixture, 50 μL ultrapure water, and 10 mg K_2CO_3 (> 99%, Merck) were added. The suspension was stirred at room temperature for 24 h. After that, the liquid phase was separated by decantation. After evaporation of the solvent to dryness under a gentle nitrogen stream using a TurboVap LV (Biotage, Uppsala, Sweden), the residue was dissolved in 1 mL ethyl acetate ($\geq 99.9\%$, VWR Chemicals, Darmstadt, Germany) and measured by GC–MS.

Methanolysis using CH_3ONa

Sodium methoxide (CH_3ONa , 0.5 M in MeOH) was obtained from Sigma-Aldrich. Exact 50 mg PET/sand mixture or sample and 10 mg IS mixture were weighed into a glass reaction vessel. DCM/MeOH mixture (5 mL) and 100 μL CH_3ONa solution were added and the suspension was stirred at room temperature. After 24 h, by adding a spatula tip of succinic acid ($\geq 99\%$, Sigma-Aldrich), the reaction was stopped and the excessive CH_3ONa was deactivated. About 1–2 mL of the supernatant was drawn up into a polypropylene syringe (Braun, Melsungen, Germany), filtered through a syringe filter (Whatman™ Spartan™, 30 mm diameter, 0.45 μm pore size; Cytiva, Little Chalfont, UK), and analyzed via GC–MS. Figure 1 shows a schematic depiction of the procedure.

GC–MS conditions

GC separation was performed on an Agilent 7890B gas chromatograph (Santa Clara, CA, USA) equipped with an ALS 7650A autosampler (Agilent) and a DB-5ms separation column (Agilent) with the dimensions: 60 m length, 0.25 mm inner diameter, 0.25 μm film thickness. The sample was injected via a programmable temperature vaporizer (PTV) injector in splitless mode with an injected volume of 1 μL and the following temperature program: 45 °C for 0.2 min, increase at $666 \text{ }^\circ\text{C min}^{-1}$ to 290 °C, hold for 10 min. The carrier gas was helium at a constant flow of 1.2 mL min^{-1} . Chromatographic separation was performed with the following temperature program: hold at 40 °C for 2 min, increase at $40 \text{ }^\circ\text{C min}^{-1}$ to 120 °C, increase at $12 \text{ }^\circ\text{C min}^{-1}$ to 240 °C, increase at $40 \text{ }^\circ\text{C min}^{-1}$ to 320 °C, hold for 5 min (total analysis time: 21 min). For detection, an MSD 5977B (Agilent) in full scan mode (m/z range 40–500, 5.9 scans s^{-1}) with electron ionization (70 eV) was used. For chromatograms, see Figures S1 and S2 in the Supplementary Information (SI).

Quantification

PET was quantified via the peak area of the methanolysis product, dimethyl terephthalate (DMT), in relation to the methanolysis product of the IS, dimethyl terephthalate- d_4 (DMT- d_4). The monitored indicator ions and retention times are listed in Table 1.

The calibration ranged from 0.001 to 1 mg g⁻¹. Calibration samples were treated and analyzed in the same way as the environmental samples. Calibration curves were fitted by the MassHunter Quantitative Analysis tool (Agilent, Santa Clara, CA, USA) using 1/x weighting.

Validation

To determine the selectivity of the developed method, the following natural, PET-free matrices were analyzed: wood and needles from a fir tree (*Abies nordmanniana*), cypress needles, oak leaves, orange peels, and organic kitchen waste. Bream filet was collected from the river Rhine in Germany and freeze-dried and milled. Humic acids (~ 30 to 40%) were purchased from Carl Roth (Karlsruhe, Germany). Exact 50 mg of matrix was analyzed, except for bream filet and humic acids, of which only 10 mg was analyzed.

To evaluate the influence of different PET sources, pellets (virgin and recycled; Carat, Bocholt, Germany), a single-use bottle, a blister, and a geogrid were ground and sieved (< 300 µm). To test applicability of the method to larger particles, the size fraction > 300 µm of the virgin pellets was analyzed, too.

Reproducibility, recovery, and limits of detection/quantification were determined using a mixture of suspended particulate matter (SPM) from the rivers Rhine, Moselle, and Saar spiked with 0.01 and 0.5 mg g⁻¹ PET.

Environmental samples

The developed analytical method was applied to a variety of environmental samples to evaluate its appropriateness. In order to reduce the risk of contamination (blank values), any contact of the samples to plastic tools and vessels was avoided during sampling and storage.

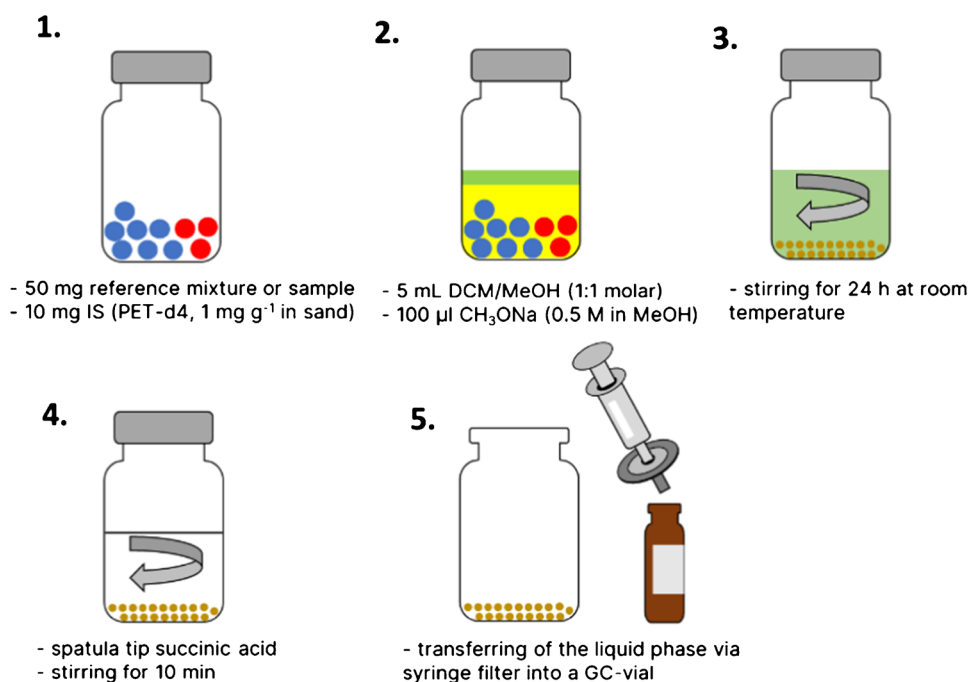
SPM from the river Rhine (km 590.3) was collected with a CEPA Z61 continuous flow centrifuge (Carl Padberg Zentrifugenbau GmbH, Lahr, Germany) in the area of Koblenz (Germany). Over a duration of 4–8 h, at least 4 m³ was sampled and at least 25 g (dry mass) matter was collected. Sediments were taken utilizing a Van Veen grab sampler or a stainless-steel spade from smaller tributaries of the river Rhine (Lahn, Nahe). Sewage sludges were taken from a wastewater treatment plant (WWTP) in Koblenz (Germany) with a population equivalent of 320,000. Not less than 500 g was collected for each sample. Compost samples were taken

Table 1 Methanolysis products and selected ions

Polymer	Methanolysis product	Indicator ions [<i>m/z</i>]	<i>t_R</i> (min)
PET	Dimethyl terephthalate	194, 163 ^a , 135	12.94
PET- d_4	Dimethyl terephthalate- d_4	198, 167 ^a , 139	12.91

^aUsed for quantification

Fig. 1 Scheme of the sample preparation including methanolysis reaction using CH₃ONa as catalyst



from composting facilities during a routine sampling by the Bundesgütegemeinschaft Kompost e. V. (BGK) using stainless-steel tools and containers. Furthermore, dust samples were collected from five different households.

All samples were ground and homogenized in a planet mill prior to analysis.

Water samples (9 L) were filtered through a glass fiber filter (MGB grade, 5 cm diameter, Sartorius). After drying, the whole filter was subjected to methanolysis prior to analysis. The following samples were taken: (a) three brands of mineral water from single-use PET bottles (6×1.5 L) from local supermarkets, (b) tap water from five different house installations in Koblenz and vicinity (Germany), and (c) ultrapure water (Milli-Q®, Merck).

Quality assurance and quality control

Only PET-free sampling devices and laboratory equipment were used. To prevent a contamination of the samples, clothing made from synthetic fibers was avoided and cotton lab coats were worn during sample preparation and analysis. All glass reaction vessels were cleaned with acetone (Pico-grade, Th. Geyer, Renningen, Germany) prior to their usage. Procedural blanks using calcined sea sand (700 °C for 3 h) were carried out during each sample series. To evaluate the risks of a sample contamination during sample preparation, opened reaction vessels were placed in the fume hood and on the lab bench where samples were prepared. After 14 days, they were analyzed as well. Furthermore, heated sea sand and ultrapure water were analyzed within each analysis batch of solid samples and water samples, respectively.

Results and discussion

The newly developed quantification method is based on complete methanolysis of the polyester PET to DMT using CH_3ONa as basic catalyst (see Fig. 2). The formed DMT is determined by GC–MS.

Method optimization and validation

Optimization of methanolysis

For optimization of the methanolysis step, different catalysts, reagent amounts, and reaction times were tested. Pham

and Cho [14] described methanolysis of PET in an equimolar mixture of DCM and MeOH using K_2CO_3 as catalyst at room temperature. This method was used as a starting point for optimization. While yields for DMT were sufficient (90–100%) at a PET concentration of 1 mg g^{-1} , yields at smaller concentrations ($< 100 \text{ } \mu\text{g g}^{-1}$) were not satisfactory (40–70%). This was caused by the formation of terephthalic acid or the monoester by hydrolysis, which was also reflected in a decrease in IS intensity.

Water-free conditions were tested to avoid this side reaction. However, without water, no depolymerization of PET took place. Apparently, a certain amount of water is needed for the activation of the catalyst K_2CO_3 . However, this challenge was solved by using sodium methoxide (CH_3ONa) as catalyst as described by Cao et al. [15]. CH_3ONa is soluble in the used solvent mixture (MeOH/DCM), which should be water-free. By this approach, recoveries $> 90\%$ were attained over a broad concentration range ($0.005\text{--}1 \text{ mg g}^{-1}$).

Since natural matrix compounds can react with the catalyst, they can hamper the efficiency of the chemolysis. In order to optimize the amount of sodium methoxide needed, recovery experiments were carried out in an authentic environmental matrix (SPM). Quantitative yields of 90–100% DMT were obtained using $100 \text{ } \mu\text{L}$ CH_3ONa (0.5 M in MeOH), whereas $50 \text{ } \mu\text{L}$ only led to conversion rates of 30–71% (data in SI, Table S1). In order to optimize the overall reaction time, the conversion rate of PET to DMT was determined after 2, 4, 8, 16, 24, and 48 h of methanolysis (see Figure S3 in SI). It was found that methanolysis was quantitative after 16 h. Due to a convenient time-scaling in the laboratory, a period of 24 h was chosen as the final reaction time.

Calibration

The preparation of different calibration points from a stock solution by serial dilution was impossible, since PET is only soluble in few solvents such as 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP), and even traces of this alcohol lead to byproducts such as methyl 1,1,1,3,3,3-hexafluoro-2-propenyl terephthalate and bis(1,1,1,3,3,3-hexafluoro-2-propenyl) terephthalate if K_2CO_3 was used as catalyst (ref. Figure S4). No chemolysis of PET was observed in the presence of NaOCH_3 and HFIP. This can be explained by higher acidity of HFIP compared to MeOH and lower nucleophilicity of the anion caused by the electron-attracting CF_3 -groups.

Fig. 2 Reaction equation of methanolysis of PET



Direct weighing of the polymers was also an unsuitable method, due to uncertainties of weighing of small quantities when preparing the lower calibration levels. Therefore, as an alternative, a serial dilution of the stock mixture was carried out with calcined sea sand, an inert matrix. This approach has already been shown to work for other MP types (PE, PP, PS) [1]. Thus, calibration points were generated by analyzing known concentrations of PET diluted with sand. A sand/PET- d_4 mixture was added as IS before start of the methanolysis. During method development, no deuterium/hydrogen exchange was observed. The deuterium labeling at the benzene ring was stable under the conditions of methanolysis. For the calibration curves, relative peak areas were calculated with regard to DMT- d_4 . A linear calibration curve over a range of 0.05–50 μg PET was achieved, which equates to concentrations of 1–1000 $\mu\text{g g}^{-1}$ for a sample weight of 50 mg (see Figure S5 in SI).

Limits of detection and quantification

The limit of detection (LOD) and limit of quantification (LOQ) were derived from the signal-to-noise ratio (SNR) in diluted PET standards (1–10 $\mu\text{g g}^{-1}$, i.e., 50–500 ng absolute PET mass), with an SNR ≥ 3 required for the LOD and an SNR ≥ 10 required for the LOQ. An LOD of 50 ng PET and an LOQ of 200 ng PET were achieved. With a typical sample in-weight of 50 mg, this equates to concentrations of 1 and 4 $\mu\text{g g}^{-1}$, respectively. For a volume of 9 L water, this equates to concentrations of 6 and 22 ng/L, respectively.

Matrix interferences

Quantification of PET is based on the indirect determination of the polymer by its methanolysis product DMT. However, DMT might also be formed from matrix ingredients present in environmental samples. Therefore, the selectivity of the method was evaluated by analyzing organic materials which were not contaminated by PET. Table 2 provides the bias for the various natural compounds. In general, the tested matrices led to a negligible bias above the LOQ, except for humic acids, which led to interferences of 0.03 mg g^{-1} . Therefore, up to a humic acid content of 10%, there is no bias above the LOQ. However, in humic acid-rich samples such as peat, an overestimation of PET cannot be excluded. Apart from that, however, the method is suitable for quantification of PET in complex environmental samples such as sediment, sewage sludge, or compost.

Contamination during sample preparation

To detect a contamination caused by atmospheric deposition [16], six opened reaction vessels were displayed for 14 days at different sites (fume hood and lab bench) in the laboratory where the sample preparation was conducted. PET levels ranged from < LOD up to 996 ng (absolute mass) or 20 $\mu\text{g g}^{-1}$ (based on a sample weight of 50 mg) (ref. Table S2). However, this was a worst-case scenario: If short exposure times were applied (few minutes compared to 14 days), the risk of contamination was found to be negligible. The background level measuring heated sea sand was always < LOQ (ref. Table S3).

Recoveries and reproducibility

To check whether different matrices have effects on the depolymerization efficiency and reproducibility, spiking experiments were performed in an SPM sample at two different concentration levels (0.01 and 0.5 mg g^{-1}). For both spike levels, recoveries of 96% were achieved. To check the reproducibility of the analytical method, the non-spiked and spiked samples were analyzed in eight replicates over the whole analytical method (shown in Table 3 and Table S4). Relative uncertainties ranged from 8 to 16%, highlighting a high reproducibility and reliability of the analytical method.

For the lower spike level, the statistical uncertainty was relatively high (36%), which was probably caused by the spiking of the authentic sediment sample, which already contained a quantity of PET in the range of the lower spike level. The statistical uncertainty was elevated in the non-spiked sample, probably caused by an inhomogeneous distribution of the PET contamination. Due to the propagation of the uncertainties of the results, the uncertainty of the recovery is elevated.

Table 2 Interferences by organic matrix

Organic material	“PET” content ($\mu\text{g g}^{-1}$)
Kitchen waste	5
Oak leaves	< 1
Fir needles	< 1
Cypress needles	< 4
Fir wood	4
Orange peel	< 1
Moss	4
Algae	< 1
Fucus	< 1
Humic acid	30
Bream filet	< 1

Table 3 Reproducibility and recoveries for PET in SPM at two spiking levels

Spiked concentration ($\mu\text{g g}^{-1}$)	Number of replicates	Average measured concentration ($\mu\text{g g}^{-1}$)	Relative uncertainty ^a (%)	Recovery average (%)
0	8	9	16	-
10	8	19	11	96 ± 36^b
500	8	490	8	96 ± 8^b

^aRelative confidence interval ($t_{n=7, p=0.95} = 2.365$); ^bcalculated by propagation of the uncertainties

Analysis of different PET products

To evaluate whether methanolysis efficiency is affected by PET type and size, several PET products were tested as well as one sample of poly(butylene adipate terephthalate) (PBAT) (Table 4). Recoveries for all PET types ranged between 87 and 117%. Even the coarse PET particle ($> 300 \mu\text{m}$) was quantitatively converted to DMT, confirming that the analytical method is not limited to the smaller size fractions. However, particles of this size could cause inhomogeneity issues and samples should be homogenized as far as possible prior to analysis.

For PBAT, a “PET” recovery of $28 \pm 16\%$ was determined (i.e., $1 \mu\text{g}$ of PBAT was transformed to an amount of DMT that would equate to $0.28 \pm 0.16 \mu\text{g}$ of PET). The partition of terephthalate in the PBAT sample was determined by NMR spectroscopy (see SI: Characterization of PBAT) to be $22 \pm 2\%$ by mass, resulting in a conversion factor for PBAT into PET of 0.29 ± 0.03 . Therefore, the recovery of the actual terephthalate content was $97 \pm 21\%$.

This experiment revealed that the developed method cannot distinguish between PET, PBAT, and other terephthalates. The determined PET concentration should therefore be understood as the sum of concentrations of all terephthalate-containing polymers (“cluster” [17]). A non-polymeric example of a terephthalate that might interfere with PET

Table 4 Recoveries for different PET products

PET product	Size fraction (μm)	Recovery average (%)
Pellet (virgin)	< 300	87 ± 19^a
Pellet (virgin)	> 300	117
Pellet (recycled)	< 300	95 ± 9^a
Bottle	< 300	76 ± 35^a
Blister	< 300	100 ± 24^a
Geogrid	< 300	77 ± 21^a
PBAT pellet	< 300	28 ± 16^a (97 ± 21^b)

^aConfidence interval ($t_{n=4, p=0.95} = 3.182$); ^bconsidering terephthalic acid content of the analyzed PBAT

Table 5 PET concentrations in environmental samples

	N	PET content
Sediments	7	$22 (< 4\text{--}68) \mu\text{g g}^{-1 a}$
Suspended particulate matter (Rhine)	21	$13 (8\text{--}37) \mu\text{g g}^{-1 a}$
Indoor dust	5	$2.4 (1.7\text{--}57) \text{mg g}^{-1 a}$
Compost	3	$27; 63; 116 \mu\text{g g}^{-1}$
Sewage sludge	6	$173 (158\text{--}231) \mu\text{g g}^{-1 a}$
Bottled water	3	$26; 106; 463 \text{ng L}^{-1}$

^aMedian (minimum–maximum)

quantification is the plasticizer di(2-ethylhexyl) terephthalate [18], which is increasingly produced as a phthalate-free plasticizer alternative. However, its production volume is significantly lower than that of PET, and thus a lower environmental concentration can be expected.

Analysis of environmental and water samples

Sediments, suspended particulate matter, compost, sewage sludge, and house dust as well as mineral water bottled in single-use PET container and tap water were analyzed with regard to their contamination with PET MP. The results are summarized in Table 5, with a detailed list in SI Tables S5–S11.

PET concentrations in SPM and sediments lie in the same range as reported for PVC ($8\text{--}220 \mu\text{g g}^{-1}$) and PP ($32\text{--}55 \mu\text{g g}^{-1}$) in previous studies, but lower than PE ($30\text{--}1000 \mu\text{g g}^{-1}$) [1, 6]. Compost and sewage sludge showed slightly higher concentrations in the same range as reported in literature [11–13].

Highest PET concentrations (up to 5.7% by mass) were found in indoor dust. High amounts of MP—especially fibers—were reported in previous studies [12, 19–21]. PET is a common synthetic fiber in textiles such as clothes and carpets. Thus, occurrence of PET in indoor dust is not surprising. However, these high amounts in indoor dust demonstrate the risk of possible sample contamination and the necessity of measures to prevent them.

Due to the enrichment of 9 L and the high sensitivity of the method, PET could be quantified in bottled waters, too. The detected PET concentrations were from equal to up to ten times as high as in tap water and ultrapure water (mean $32 \pm 15 \text{ng/L}$), which were used as blank controls. Contamination of bottled water by microplastics was reported in previous studies [22–25]. However, measured concentrations were much lower than reported in literature (0.2 to $6 \mu\text{g L}^{-1}$ [26] and 100 to $3000 \mu\text{g L}^{-1}$ [27]).

Conclusion and outlook

A quantification method for PET in different environmental samples was developed and validated. Using CH_3ONa as catalyst, complete methanolysis of PET to DMT was possible even in complex matrices such as sewage sludge. The newly developed method enables fast and straightforward quantification of PET. Using PET- d_4 as IS allows for reliable quantification results. However, since it is an indirect determination of PET via its methanolysis product DMT, other compounds forming DMT during methanolysis can disturb the quantification of PET and lead to overdeterminations. Various tested natural compounds were tested and led to negligible signals. Synthetic polymers containing terephthalic acid such as PBAT, polybutylene terephthalate (PBT), or glycol-modified PET (PETG) or terephthalic acid esters used as phthalate-free plasticizer alternatives with increasing production volumes are potential DMT precursors. Consequently, results have to be seen as a sum of all terephthalic acid-containing compounds expressed as PET. However, PET is by far the most important synthetic terephthalic acid derivative.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00216-025-05963-4>.

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Data availability All data generated or analyzed during this study are included in the published article and its supplementary materials.

Declarations

Conflict of interest The authors declare no competing interests.

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